

PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Date	12 June 2026	Team ID	
Project Name	E-commerce Application (ShopEZ)	Maximum Marks	4 Marks

TECHNICAL ARCHITECTURE

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2
Guidelines: Include all processes, local/cloud boundaries, external interfaces, data storage, and environment tools.

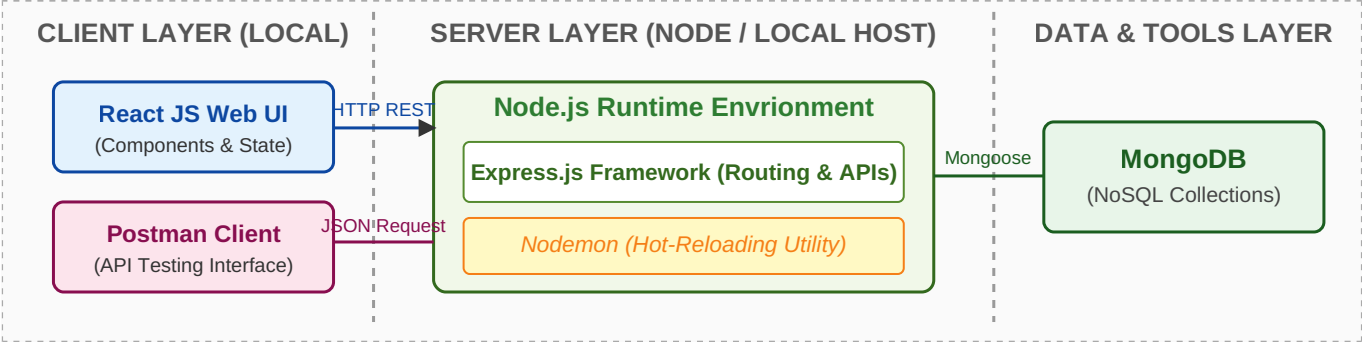


TABLE-1 : COMPONENTS & TECHNOLOGIES

S.NO	COMPONENT	DESCRIPTION	TECHNOLOGY
1	User Interface	Responsive single-page web interface for browsing items, modifying carts, checking out, and accessing admin views.	React JS, HTML5, CSS3, JavaScript (ES6)
2	Application Logic-1	Backend processing server handling client requests, routing, user authentication, inventory validation, and business workflow.	Node.js, Express.js Framework
3	Application Logic-2	Object-Data Modeling (ODM) layer implemented within application server logic to structure schemas and interact with database collections.	Mongoose Library
4	Application Logic-3	Development-stage automatic script reloading utility tracking code alterations without requiring manual runtime resets.	Nodemon Utility
5	Database	Local structural NoSQL data repository storing JSON-like document datasets for users, catalog inventories, logs, and purchase files.	MongoDB (Local Instance)
6	Cloud Database	Not applicable for current architectural iteration; standard MongoDB deployment localized to host system resources.	None (Localized Data Stack)
7	File Storage	Storage allocation for static storefront multimedia, asset binaries, and uploaded catalog visual data directly on server volumes.	Local File System Storage
8	External API-1	Independent interface application client used by developers to manually execute, debug, and validate backend REST route integrity.	Postman API Client Tool
9	External API-2	External computational integrations or mock validation modules utilized during transaction checkout routines.	Mock Payment API System
10	Machine Learning Model	Not integrated into core structural application parameters for this baseline implementation scope.	None
11	Infrastructure (Server)		

S.NO	COMPONENT	DESCRIPTION	TECHNOLOGY
		Host environment running localized execution runtimes for client views and Node servers during development cycles.	Localhost, Node Runtime Environment

TABLE-2: APPLICATION CHARACTERISTICS

S.NO	CHARACTERISTICS	DESCRIPTION	TECHNOLOGY
1	Open-Source Frameworks	Leverages fully open-source MERN ecosystem components to manage dynamic data binding and asynchronous server event loops.	React JS, Express.js, Node.js
2	Security Implementations	Secures operational interfaces via encrypted stateless browser cookies, protected API endpoints, and credential hash procedures.	JSON Web Tokens (JWT), Bcrypt
3	Scalable Architecture	Maintains separate decoupling of presentation tiers (React) and application logical controllers (Express) to enable horizontal scaling.	Decoupled MERN 3-Tier Pattern
4	Availability	Ensured locally through light multi-thread sub-processes and standard Node error handling callbacks preventing app termination.	Node.js Process Cluster Bounds
5	Performance	Optimized via non-blocking asynchronous event I/O cycles, single-page DOM rendering, and optimized index tracking in collections.	MongoDB Indexing, Async-Await

REFERENCES

<https://c4model.com/>
<https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>
<https://www.ibm.com/cloud/architecture>
<https://aws.amazon.com/architecture>
<https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>